

Graph Neural Networks for Street-Level Flood Forecasting in Dense Tropical Cities

Nadia Putri Rahmawati^{1*}, Bagus Adi Nugroho¹, Siti Aminah²

¹ MGM Laboratory, Department of Informatics

² Faculty of Computer Science, a university in East Java, Indonesia

* Corresponding author

ABSTRACT

Flash floods in dense tropical cities rise and fall within an hour, which is faster than most hydrological models can be recalibrated. We model the drainage network of a mid-sized Indonesian city as a graph of 4,812 street segments and train a spatio-temporal graph neural network on three rainy seasons of gauge, radar and crowd-reported inundation data. The model predicts water depth per segment 30 to 90 minutes ahead. On a held-out season it reduces mean absolute error by 23 percent compared with a gridded convolutional baseline and flags 81 percent of the flooded segments at least 45 minutes in advance. We release the graph construction code and an anonymized benchmark.

Keywords: flood forecasting, graph neural networks, urban drainage, rainfall nowcasting

1. Introduction

Every rainy season, short and intense storms turn a handful of streets into rivers. Residents know which corners flood first, but city dashboards usually show rainfall, not consequences. The gap between a radar image and a flooded street is exactly where a street-level forecast is useful.

Physically based models capture this well but need detailed channel geometry and slow calibration. Data-driven models are fast but often ignore how water actually moves along the network. We try to keep the best of both by letting the drainage topology shape the model itself [1, 2].

2. Method

We build a directed graph where nodes are street segments and edges follow the dominant drainage direction derived from a 1 m digital elevation model. Each node carries static features (slope, impervious fraction, distance to the nearest channel) and dynamic features (radar rainfall over the last three hours, upstream gauge levels).

The forecaster stacks three graph attention layers with a gated temporal convolution in between. Training uses a depth-weighted loss so that the rare deep floods matter more than the many dry segments. We train on two seasons and test on the third, with no spatial leakage between splits.

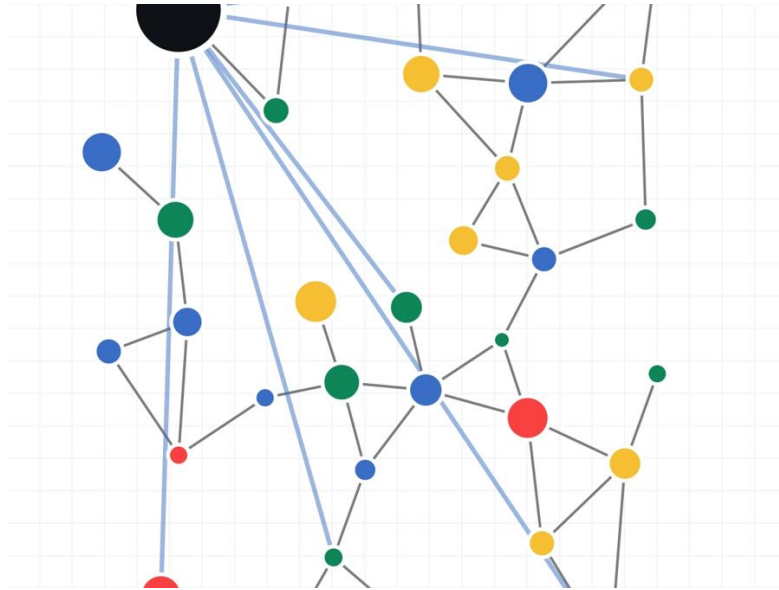


Figure 1. The drainage graph. Node size shows upstream catchment area and highlighted edges show the main flow paths into the lowest segment.

3. Results

The graph model beats both the persistence baseline and the gridded convolutional network on all lead times. Gains grow with lead time, which suggests the topology helps the model reason about water that has not arrived yet (Table 1). Most remaining errors come from blocked culverts, which no input feature can see.

Model	MAE 30 min	MAE 60 min	Recall at 45 min
Persistence	7.9 cm	12.4 cm	0.42
Gridded CNN	5.1 cm	8.8 cm	0.69
Graph model (ours)	4.2 cm	6.6 cm	0.81

Table 1. Water depth error and early warning recall on the held-out season.

4. Discussion

Crowd reports are noisy but surprisingly valuable: removing them costs 6 points of recall. We also found that a simple rule that suppresses alerts on segments with fewer than three historical floods cuts false alarms by a third without hurting recall much.

5. Conclusion

Letting the drainage graph structure the network is a cheap, strong prior for flood forecasting. Next we want to test transfer to a second city with only one season of labels.

References

[1] Kipf, T. N. and Welling, M. Semi-supervised classification with graph convolutional networks. ICLR, 2017.

[2] Velickovic, P. et al. Graph attention networks. ICLR, 2018.

[3] Wu, Z. et al. Graph WaveNet for deep spatial-temporal graph modeling. IJCAI, 2019.

[4] Rahmawati, N. P. Crowd reports as ground truth for urban inundation. Zemi working notes, 2025.

[5] Nugroho, B. A. and Aminah, S. Building drainage graphs from open elevation data. Preprint, 2025.

[6] Bates, P. D. et al. A simple inertial formulation of the shallow water equations. Journal of Hydrology, 2010.